

Exercise 4b

Deadline: 08.01.2026 22:00.

Ask questions in discord to #ask-your-tutor

In this exercise, you will familiarize yourselves with simulation-based inference, using the SIR equations from epidemiology as an example.

Regulations

Please implement your solutions in form of *Jupyter notebooks* (*.ipynb files), which can mix executable code, figures and text in a single file. They can be created, edited, and executed in the web browser, in the stand-alone app JupyterLab, or in the cloud via Google Colab and similar services.

Create a Jupyter notebook `epidemiology-part-2.ipynb` for your solution and export the notebook to HTML as `epidemiology-part-2.html`. Zip all files into a single archive `ex04b.zip` and upload this file to MaMPF before the given deadline.

Moreover, please set your **Anzeigename/display name** and **Name in Uebungsgruppen/name in tutorials** in MaMPF to your real name, which should be identical to your name in `muesli` and make sure you **join the submission** of your team via the invitation code before the submission deadline. Check out <https://mampf.blog/handing-in-homework-assignments> for instructions.

1 Validation

Determine the quality of the INN from exercise 04a in terms of posterior calibration. For the model and training, you can just copy the relevant code into your new notebook, possibly with improvements. Then check that the ground truth values are uniformly distributed in sorted samples from both the marginal distributions of the individual parameters $\hat{\lambda}$, $\hat{\mu}$, and $\hat{I}_0$, and from the distribution of “surprisals” $E(Y) = -\log p(Y|h(X))$ of the joint prediction $\hat{Y}$. To this end, for multiple fixed X_i , create samples $\{\hat{Y}_k \sim p(Y|h(X_i))\}_{k=1}^M$ of parameter estimates from the posterior, and use both the histogram and the empirical CDF methods as explained in the lecture.

Try to improve your models and training procedures until the calibration tests succeed.

2 Noisy data

We now extend the basic SIR model with a noise model for the observations. Implement the new function `simulate_sir_noisy(lam, mu, I_0, L, rho, sigma_2, T)`, which outputs noisy sequences

$$\tilde{X} = [(\Delta\tilde{S}(1), \Delta\tilde{R}(1)), \dots, (\Delta\tilde{S}(T), \Delta\tilde{R}(T))]$$

We assume that the noise is multiplicative Gaussian, i.e. follows the rule

$$\begin{aligned} \Delta\tilde{S}(t) &= \Delta S(t-L) \cdot \epsilon_S & \text{with} & & \epsilon_S &\sim \mathcal{N}(\rho, \sigma^2) \\ \Delta\tilde{R}(t) &= \Delta R(t-L) \cdot \epsilon_R & \text{with} & & \epsilon_R &\sim \mathcal{N}(\rho, \sigma^2) \end{aligned}$$

The new parameters $L \geq 0$, $\rho \leq 1$, and σ^2 represent the reporting delay, underreporting fraction, and reporting error variance respectively. Since ϵ_S and ϵ_R must be non-negative, reject negative outputs from the Gaussians and draw new random numbers until you get a positive sample (“rejection sampling”).

Visualize the difference between noise-free outcomes X and noisy outcomes $\tilde{X}$ to find good ranges for the new parameters and extend the prior $p^{\text{sim}}(Y)$ accordingly.

Extend your networks from exercise 04a to accomodate the additional parameters of the inverse problem. Repeat all experiments (i.e. point estimation and posterior estimation as in exercise 04a) and validity checks (posterior plots and resimulations as in exercise 04a, and calibration checks as in task 1) with the stochastic simulation and comment on your findings.

3 Model misspecification detection

We now repeat the model misspecification check with data that are clearly misspecified: We use the summary network from task 4 of exercise 04a, which was trained with noise-free data, but test it with data from the noisy simulation. The χ^2 and Kolmogorov-Smirnov tests should now clearly signal that noisy data are beyond the network's training domain and should not be processed because results will not be trustworthy. Conversely, data from a noise-free simulation should be accepted.

When you have difficulty shaping the summary distribution into a standard normal by the MMD loss (i.e. the trade-off between MMD regularization and inverse inference accuracy is unfavourable), you can alternatively try ex-post training: After normal training of the summary network with noise-free data, you get some unknown summary distribution $p(\hat{h}(x))$ (it may still be beneficial to use weak MMD regularization to prevent $p(\hat{h}(x))$ from becoming very complicated). Then pass noise-free validation data through the summary network to obtain a sample $\{s_i \sim p(\hat{h}(x))\}$ and use it to train an additional normalizing flow which approximates $\hat{p}(\hat{h}(x)) \approx p(\hat{h}(x))$. Then perform the χ^2 and Kolmogorov-Smirnov tests in the latent space of this normalizing flow— for inlier data, this latent distribution should be very close to standard normal.